JOSE RIZAL UNIVERSITY
COLLEGE OF COMPUTER STUDIES AND ENGINEERING
INFORMATION TECHNOLOGY DEPARTMENT

VISION

JRU will be a market leader in the use of technology for innovation in teaching and learning to produce graduates of social importance.

MISSION

The University aims to develop its students to become useful and responsible citizens through the effective transfer of relevant knowledge and desirable values.

INSTITUTIONAL OUTCOME

A JRU graduate has the competencies and values in the disciplinal area completed such that he/she is a useful and responsible citizen of the country.

JRU GRADUATE ATTRIBUTES

A JRU graduate is equipped with the qualities and competencies to excel in their profession, contribute to society, and thrive in a changing world. Specifically, he/she is a/an:
GA1: Critical and creative thinker Can apply logical and disciplined process of rationalizing, analyzing, evaluating, and interpreting information to conceive innovative ideas or solutions.
GA2: Effective Communicator and collaborator Uses diverse communication platforms to effectively engage with others in a variety of roles and interests to build shared understanding and achieve outcomes productively and efficiently
GA3: Lifelong Learner Engages in reflective learning focused on continuous professional and personal improvement.
GA4: Ethical and Responsible Citizen Practices high ethical standards of work driven by a strong sense of personal and social responsibility to perform works and services that provide solutions to the problems in society.
GA5: Professionally Competent Demonstrate broad professional knowledge, attitudes, and skills required to deliver effective work performance in various workspaces in the organization.
GA6: Technology-enabled Professionals Utilizes suitable ICT tools and technologies as a practicing professional resilient to any industry and technological advances.

Course Title: **Database Fundamentals**
Course Code: ITC C201
Credit Units: 3 units
Contact Hours: 2 hours' lecture, 3 hours' laboratory
Course Description: This course introduces database concepts, database design, creation, developments, use, and management. It emphasized on data dictionaries, normalization, data integrity, data modelling and creation of simple tables, queries, reports, and forms. Using Oracle Database Basic SQL statements and the rules such as data definition, data manipulation, and data query statements are discussed. This course culminates with a project that challenges students to design, implement, and demonstrate a normalized database structure by creating simple database tables, queries, reports, and forms as a database solution for a business organization.

Placement: Second Year, 1st semester
Pre-Requisite: MAT C604

Program Educational Objectives (PEO)	**Graduate Attributes**					
Three to five years after completing the BS Information Technology program, the graduates will:	**GA1 - Critical & Creative Thinker**	**GA2 0- Effective Communicator & Collaborator**	**GA3 - Lifelong Learner**	**GA4 - Ethical & Responsible Citizen**	**GA5 - Professionally Competent**	**GA6 - Technologically Enabled Professional**
1. Excel in their profession/career utilizing the knowledge acquired in the BS InformatinTechnology program;					√	
2. Become effective collaborators and innovators in hardware and/or software-related computer engineering projects, applying professional/technical skills and competencies to make a positive impact on society; and	√	√		√		√
3. Be engaged in life-long learning and professional development through self-study, continuing education, or graduate and professional studies.			√			

Student Outcomes	**Program Educational Objectives (PEO)**		
By the time of graduation, students of BS Information Technology will have the ability to:	**PEO1**	**PEO2**	**PEO3**
a. Apply the knowledge of computing, science and mathematics appropriate to the discipline.	√		
b. Understand best practices and standards and their applications.	√		
c. Analyze complex problems and identify and define the computing requirements appropriate to its solution.		√	
d. Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.		√	
e. Design, implement and evaluate computer-based systems, processes, components or programs to meet desired needs and requirements under various constraints.		√	
f. Integrate IT-based solutions into the outer environment effectively.		√	
g. Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT position.	√		
h. Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.		√	
i. Assist in the creation of effective IT project plan.			
j. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		√	√
k. Analyze the local and global impact of computing information technology on individuals, organizations, and society.	√		
l. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.	√		
m. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.			√

Relationship of Course Learning Outcomes to Student Outcomes

Course Learning Outcomes (CLO)	**Student Outcomes**												
At the end of the course, students are expected to:	**a**	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**	**m**
CLO1 Define the concepts and applications of database, different database models, database management systems, relational database theory;	√		√	√									
CLO2 - Identify and apply data modelling concepts and their application in design and development process based on a situational synthesis of a business problem; convert the logical model into a physical model and apply normalization in designing a database.	√		√	√		√				√			
CLO3. List and use basic SQL statement to create, manipulate, and query database; and construct one working system using relational database and a front-end language based on the requirements provided by the faculty	√		√	√								√	

.

Course Outline:

Time Frame	Topic/Content	Course Learning Outcomes	Intended Learning Objectives	Teaching/learning Activities	Assessment Task	Indicator
Week 1	1. Class Orientation 1.1 JRU Mission, Vision, Core values 1.2 Orientation to Outcomes-Based Education, Institutional Outcomes, Program Educational Objectives, Program Outcomes and Course Outcomes 1.3 Orientation Classroom Requirements and Laboratory Guidelines, Academic Integrity, and Grading System 1.4 Sustainability and UN SDG's	Understand the concept of outcomes-based education and infer the results and applications of the learning acquired in the course.	Familiarize the students with JRU and classroom policies and grading system Discuss and enumerate the course requirements Learn the different UN SDGs and how it can be aligned to the course and program	Collaborative discussion and	Class participation	100% Class participation
Week 1–2	DATABASE Fundamentals DATABASE FOUNDATION - What is Database - Types of Databases - Evolution of File System Data Processing - Types of Data Anomaly - Components of Database - Environment DBMS Functions - Disadvantage of Database System Data Modeling and Data Models Business Rules	CLO1 CLO3	Compare Data vs Information Identify types of Data Describe Components of Database Describe the components of Database, its Advantages and Disadvantages. Identify Types of Data Models and its uses.	Collaborative discussion and interactions Interactive Learning Activity1 Identify types of data Prelim Assignment No.1- Types of DBMS Types of Data Models	Prelim Lecture activity output / Recitation (PT1.1) Prelim Quiz 1 (WW1)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Seat work & Laboratory Activity Rubrics on Seat work and Laboratory Activity

	Types of DBMS Types of Data Models					
Week 3 -4	Entity Relationship Modeling and ERDs Basic Notation of ER-Diagram Entities, Attributes, and Relationship Classification of Attributes and Identifiers Identifying Relationships and Cardinalities Enhanced Entity Relationships Modeling SuperTypes Subtypes Generalization Specialization Completeness Constraints Disjointness Constraints Subtype Discriminator ORACLE SQL DATA MODELER Logical Design and Relational Design	CLO1 CLO3 CLO2	Use Oracle Application Express Construct Logical and Physical Model Using SQL Data Modeler Design ERD Describe Classification of Attributes and Identifiers	Prelim Lecture activity output/ Recitation Collaborative discussion and interactions Interactive Learning Activity Demonstration ORACLE SQL DATA MODELER Seatwork: Analysis Business Scenario using EER-D new notation	Laboratory Activity1: Creation Logical and Relational Design (PT1.2) Prelim Quiz 2 (WW2)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Quizzes , SW & Laboratory Activity Rubrics on SW/ ASSIGNMENT/ Laboratory Activity
WEEK 5	PRELIM TERM PROJECT Identify a problem domain in any of the UN SDGs 4, 7, 8, 9 that can be applied in				Individual Prelim Project (P1)	70% of the students must get a rating <= 3.0 on the 5-point rating scale

	Database Case Scenario Analysis					Rubrics on Project /Laboratory Activity

Week 6	Summative Assessment	CLO3	Demonstrate and apply learning from previous lessons.	Self-Check: Review for Prelims	Prelim Examination	70% must get <=3.0 in 5-point rating scale
WEEK 7	CREATING ORACLE APEX ACCOUNT - Creating Oracle Database and Tables	CLO1 CLO2	Creating Oracle APEX account Creating Oracle Database and Tables	Demonstration of Oracle Application Express Assignment: Creating ORACLE APEX Account	Laboratory Activity: Creating Oracle Database and Tables (PT2.1)	
WEEK 8	Normalization of Database Tables First Normal Form Second Normal Form Third Normal Form	CLO2	Describe Database Normalization Construct a Normalization Table	Collaborative discussion and Interactions Interactive Assignment: Normalize the given table up to Third Normal Form	Lecture Activity Output/ Recitation (PT2.2)	70% of the students must get a rating <= 3.0 on the 5-point rating Rubrics on Activity
WEEK 9	ORACLE SQL - MODIFYING TABLES - DELETING TABLES - SETTING SQL CONSTRAINTS PRIMARY KEY, NOT		Modifying Tables using Alter & Drop SQL Statements Set SQL Constraints		Laboratory Activity: Modifying Tables and Setting SQL Constraints	70% of the students must get a rating <= 3.0 on the 5-point Rating

	NULL, UNIQUE, CHECK, FOREIGN KEY				(PT2.3) Midterm Quiz No.1 - DDL (WW3)	Rubrics on Activity
WEEK 10	ORACLE SQL - Insert Statement - -Update Statement - Delete Statement	CLO1 CLO2	Construct SQL Insert, Update and Delete SQL statements.	Collaborative discussion and interactions	Laboratory Activity: DML Inserting Records (PT2.4) Midterm Quiz No.2 - SQL DML Statements (WW4) Laboratory Activity: DML Updating and Deleting Records (PT2.5)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Laboratory Activity and Quizzes Rubrics on Laboratory Activity

WEEK 11	MID TERM PROJECT Identify a problem domain in any of the UN SDGs 4, 7, 8, 9 that can be applied in Database Case Scenario Analysis	CLO3	Submit written project documentation Design and present a technology-based policy solution that promotes to build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation (SDG 9		Individual Midterm Project (P2)	70% of the students must get a rating <= 3.0 on the 5-point rating scale Rubrics on Project /Laboratory Activity
Week 12	Summative Assessment	CLO3	Demonstrate and	Self-Check: Review	MIDTERM	70% must get

			apply learning from previous lessons.	for Midterms	Examination	<=3.0 in 5-point rating scale

WEEK 13-14	SQL OPERATORS, WHERE, AND, OR, NOT SQL ALIAS, LIKE, WILDCARDS, IN, BETWEEN	CLO1 CLO2	Execute Oracle Database Program Using SQL Where, And, Or, Not, Alias, Wildcard, Between & In Statements	Collaborative discussion and interactions	Laboratory Activity: SQL OPERATORS (PT3.1) Final Quiz 1 (WW5)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Quizzes
Week 15 -16	Group functions, Inner versus Outer Joins, Self Joins	CLO2 CLO3	Describe SQL Joins, and Group Functions Execute SQL Group Functions, Cross Joins, and Natural Joins Execute Inner, Outer & Self Joins	Demonstration SQL Statement in Oracle APEX	Laboratory Activity: Joining Tables (PT3.2) Final Quiz # 2 (WW6)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Seat works and Quizzes Rubrics on Seat works

Week 17	FINAL GROUP PROJECT E-Commerce Application with database connectivity Collaboration with OOP Programing Subject Group Project Study Presentation FINAL TERM PROJECT Identify a problem domain in any of the UN SDGs 4, 7, 8, 9, 11. 12 that can be applied in Database Design Analysis and E-Commerce Web development	CLO3	Submit written project documentation and conduct oral presentations. Design and present a technology-based policy solution that promotes to build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation (SDG 9)	Group Project Study Presentation	Submission of e-Portfolio (P3.1) Group Project Submission and Presentation (P3.2)	70% of the students must get a rating <= 3.0 on the 5-point rating scale on Group Project Presentation Rubrics on Group Project Presentation 100% project submission
Week 18	Summative Assessment	CLO3	Demonstrate understanding of all topics covered.	Self-Check: Review for Finals	Final Term Examination	70% of the class must get <=3.0 in 5-point rating scale Rubrics on e-Portfolio

Course Requirements:

a. Quizzes
b. Major Examination
c. Case Analysis & Laboratory Exercises
d. Discussion boards
e. Portfolio/Research/Project

Grading System and Components

Prelim Grade	**30%**	**Class standing**	**70%**	***Written Works (Quizzes/) Seatwork***	***25%***	***Written work 1***	***CLO1-WW1***	***12.50%***
						Written work 2	***CLO1-WW2***	***12.50%***
				Performance Tasks (Lab Activity)Assignments	***25%***	***Performance task***	***CLO2-PT1***	***25.00%***
				Product (Project)	***20%***	***Product 1***	***CLO3-P1***	***20.00%***
		Major Exam	**30%**			***Prelim Exam***	***CLO1, CLO2, CLO3-PE***	***30.00%***
Midterm Grade	**30%**	**Class standing**	**70%**	***Written Works (Quizzes/Assignments)***	***25%***	***Written work 1***	***CLO1-WW3***	***12.50%***
						Written work 2	***CLO1-WW4***	***12.50%***
				Performance Tasks (Lab Activity)	***25%***	***Performance task***	***CLO2-PT2***	***25.00%***
				Product (Project)	***20%***	***Product 1***	***CLO3-P2***	***20.00%***
		Major Exam	***30%***			***Midterm Exam***	***CLO1, CLO2, CLO3-ME***	***30.00%***

Final Term Grade	**40%**	**Class standing**	**70%**	***Written Works (Quizzes/Assignments)***	***25%***	***Written work 1***	***CLO1-WW5***	***12.50%***
						Written work 2	***CLO1-WW6***	***12.50%***
				Performance Tasks (Lab Activity)	***20%***	***Performance task***	***CLO2-PT3***	***12.50%***
				Product (Project)	***20%***	***Product 1***	***CLO3-P3***	***20.00%***
		Major Exam	**30%**			***Final Exam***	***CLO1, CLO2, CLO3-FE***	***30.00%***
FINAL GRADE	**100%**							

Textbook: Kuzmiakova, A. (Ed.) (2023). The Creation and Management of Database Systems. Arcler Press.https://library.jru.edu/cgi-bin/koha/opac-retrieve-file.pl?id=6e5a9af536b9a6027651ab5fcefde7de (suggested reading) e-book

References:

- Duckett, J., Ullman, C., Stone, E. (2022). *PHP & MySQL : server-side web development*. John Wiley & Sons, Inc. (Accession No.0126737)
- Salter, K. (2024).Fundamentals of Database Systems. Toronto Academic Press. (Accession No. 0128670)
- Uckun, F. (2024) .Core concepts in computer science : database.3G E-learning. (Accession No. 0129580)
- Weiss K. A., Balti H., . (2023). Job ready SQL. John Wiley & Sons, Inc. (Accession No. 0128328)

Online Reference:

- Oracle. (2023). Introduction to Oracle Database. Oracle Database Documentation. (https://docs.oracle.com/en/database/oracle/oracle-database/18/cncpt/introduction-to-oracle-database.html#GUID-A42A6EF0-20F8-4F4B-AFF7-09C100AE581E)
- Oracle. (1994-2023). Oracle Data Access. Oracle Database Documentation. (https://docs.oracle.com/cd/E56485_01/index.htm)
- W3Schools. (1999-2023). SQL Tutorial. W3Schools SQL Tutorial. (https://www.w3schools.com/sql/)

- Tutorials Point (India) Private Limited. (2023). Learn SQL. Tutorials Point SQL. (https://www.tutorialspoint.com/sql/index.htm)
- Microsoft. (2022). Open SQL Server database by using SQL Server .NET Data Provider with Visual Basic .NET. Microsoft Docs. (https://learn.microsoft.com/en-us/troubleshoot/developer/dotnet/framework/general/open-database-by-sql-server-dotnet-data-provider)

Prepared By:	Reviewed By:	Adopted textbook and references reviewed by:	Recommending Approval	Approved By	Date of Effectivity
 Virginia B. Loyola	 Dan Jayson Panergo	 Rechy Llamas	 Sharmaine Justyne R. Maglapuz	 Eng. Liza R. Reyes	SY 2026-2027
Faculty Member	OBE Team	Higher Education Libraries	Chair Person	Dean	